

Лабораторная работа №16

Дисциплина: Администрирование сетевых подсистем

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1. Цель работы
2. Выполнение лабораторной работы
3. Выводы

Раздел 1

1. Цель работы

1.1 Цель работы

- Получение навыков работы с Fail2ban

1.1 Цель работы

- Получение навыков работы с Fail2ban
- Настройка базовой защиты от атак типа brute force

Раздел 2

2. Выполнение лабораторной работы

2.1 Защита с помощью Fail2ban

2.1.1 Установка Fail2ban

- Установлен пакет fail2ban через dnf

```
[root@server.bahis.net ~]# dnf -y install fail2ban
Last metadata expiration check: 0:09:36 ago on Fri 13 Feb 2026 12:57:08 PM UTC.
Dependencies resolved.
=====
Package                        Architecture    Version         Repository      Size
=====
Installing:
fail2ban                      noarch         1.1.0-6.el9     epel            9.3 k
Installing dependencies:
fail2ban-firewalld           noarch         1.1.0-6.el9     epel            9.5 k
fail2ban-selinux              noarch         1.1.0-6.el9     epel            31 k
fail2ban-sendmail             noarch         1.1.0-6.el9     epel            12 k
fail2ban-server               noarch         1.1.0-6.el9     epel            465 k
Transaction Summary
=====
Install 5 Packages
```

2.1 Защита с помощью Fail2ban

2.1.1 Установка Fail2ban

- Установлен пакет fail2ban через dnf
- Установлены зависимости: fail2ban-firewalld, fail2ban-selinux, fail2ban-sendmail, fail2ban-server

```
[root@server.bahis.net ~]# dnf -y install fail2ban
Last metadata expiration check: 0:09:36 ago on Fri 13 Feb 2026 12:57:08 PM UTC.
Dependencies resolved.
=====
Package                        Architecture    Version         Repository      Size
=====
Installing:
fail2ban                      noarch         1.1.0-6.el9    epel            9.3 k
Installing dependencies:
fail2ban-firewalld           noarch         1.1.0-6.el9    epel            9.5 k
fail2ban-selinux              noarch         1.1.0-6.el9    epel            31 k
fail2ban-sendmail             noarch         1.1.0-6.el9    epel            12 k
fail2ban-server               noarch         1.1.0-6.el9    epel            465 k
Transaction Summary
=====
Install 5 Packages
```


2.1 Защита с помощью Fail2ban

2.1.1 Установка Fail2ban

- Установлен пакет fail2ban через dnf
- Установлены зависимости: fail2ban-firewalld, fail2ban-selinux, fail2ban-sendmail, fail2ban-server
- Использован репозиторий epel

```
[root@server.bahis.net ~]# dnf -y install fail2ban
Last metadata expiration check: 0:09:36 ago on Fri 13 Feb 2026 12:57:08 PM UTC.
Dependencies resolved.
=====
Package                Architecture    Version         Repository      Size
=====
Installing:
fail2ban                noarch          1.1.0-6.el9     epel             9.3 k
Installing dependencies:
fail2ban-firewalld      noarch          1.1.0-6.el9     epel             9.5 k
fail2ban-selinux        noarch          1.1.0-6.el9     epel             31 k
fail2ban-sendmail       noarch          1.1.0-6.el9     epel             12 k
fail2ban-server         noarch          1.1.0-6.el9     epel            465 k
Transaction Summary
=====
Install 5 Packages
```

2.1.2 Запуск и добавление службы в автозагрузку

- Выполнен `systemctl start fail2ban`

```
root@server.bahis.net ~]# systemctl start fail2ban
root@server.bahis.net ~]# systemctl enable fail2ban
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service → /usr/lib/systemd/system/fail2ban.service.
root@server.bahis.net ~]#
```

Рисунок 2: Запуск службы fail2ban и включение автозагрузки

2.1.2 Запуск и добавление службы в автозагрузку

- Выполнен `systemctl start fail2ban`
- Выполнен `systemctl enable fail2ban`

```
root@server.bahis.net ~]# systemctl start fail2ban
root@server.bahis.net ~]# systemctl enable fail2ban
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service → /usr/lib/systemd/system/fail2ban.service.
root@server.bahis.net ~]#
```

Рисунок 2: Запуск службы fail2ban и включение автозагрузки

2.1.2 Запуск и добавление службы в автозагрузку

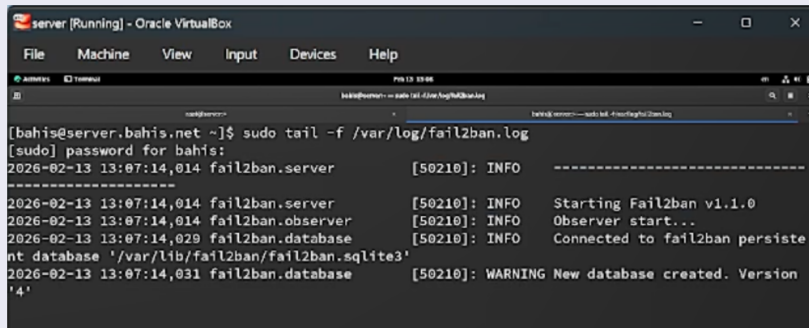
- Выполнен `systemctl start fail2ban`
- Выполнен `systemctl enable fail2ban`
- Создана ссылка в `multi-user.target.wants`

```
root@server.bahis.net ~]# systemctl start fail2ban
root@server.bahis.net ~]# systemctl enable fail2ban
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service → /usr/lib/systemd/system/fail2ban.service.
root@server.bahis.net ~]#
```

Рисунок 2: Запуск службы fail2ban и включение автозагрузки

2.1.3 Контроль журнала событий Fail2ban

- Выполнен `tail -f /var/log/fail2ban.log`



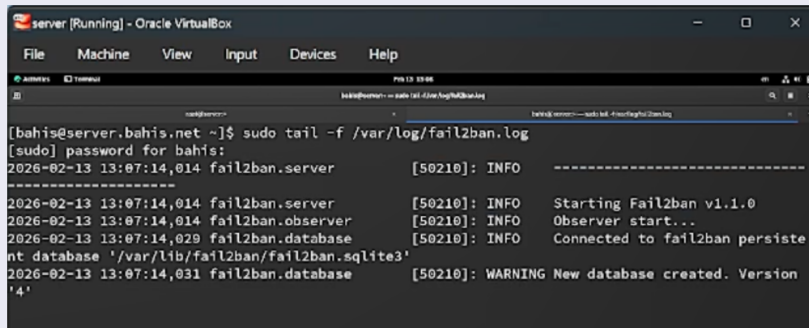
The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal displays the command `sudo tail -f /var/log/fail2ban.log` being executed. The output shows the Fail2ban service starting, including messages about the observer starting and the database being created. The log entries are as follows:

```
[bahis@server.bahis.net ~]$ sudo tail -f /var/log/fail2ban.log
[sudo] password for bahis:
2026-02-13 13:07:14,014 fail2ban.server [50210]: INFO -----
-----
2026-02-13 13:07:14,014 fail2ban.server [50210]: INFO Starting Fail2ban v1.1.0
2026-02-13 13:07:14,014 fail2ban.observer [50210]: INFO Observer start...
2026-02-13 13:07:14,029 fail2ban.database [50210]: INFO Connected to fail2ban persis
nt database '/var/lib/fail2ban/fail2ban.sqlite3'
2026-02-13 13:07:14,031 fail2ban.database [50210]: WARNING New database created. Version
'4'
```

Рисунок 3: Просмотр журнала fail2ban.log в реальном времени

2.1.3 Контроль журнала событий Fail2ban

- Выполнен `tail -f /var/log/fail2ban.log`
- Зафиксирован запуск Fail2ban 1.1.0



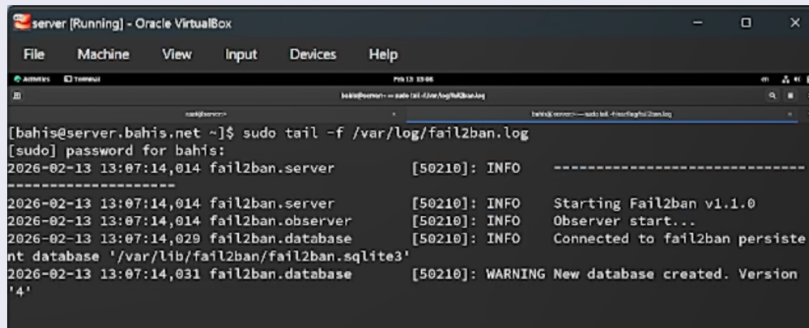
The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal output displays the command `sudo tail -f /var/log/fail2ban.log` being executed. The log shows the following entries:

```
[bahis@server.bahis.net ~]$ sudo tail -f /var/log/fail2ban.log
[sudo] password for bahis:
2026-02-13 13:07:14,014 fail2ban.server [50210]: INFO -----
-----
2026-02-13 13:07:14,014 fail2ban.server [50210]: INFO Starting Fail2ban v1.1.0
2026-02-13 13:07:14,014 fail2ban.observer [50210]: INFO Observer start...
2026-02-13 13:07:14,029 fail2ban.database [50210]: INFO Connected to fail2ban persis
nt database '/var/lib/fail2ban/fail2ban.sqlite3'
2026-02-13 13:07:14,031 fail2ban.database [50210]: WARNING New database created. Version
'4'
```

Рисунок 3: Просмотр журнала fail2ban.log в реальном времени

2.1.3 Контроль журнала событий Fail2ban

- Выполнен `tail -f /var/log/fail2ban.log`
- Зафиксирован запуск Fail2ban 1.1.0
- Инициализирован observer



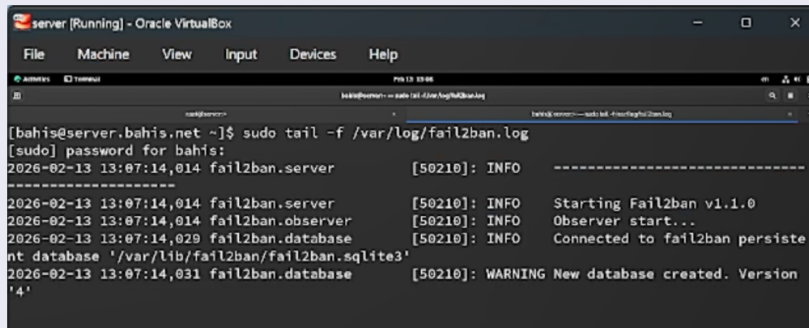
The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal displays the output of the command `sudo tail -f /var/log/fail2ban.log`. The output shows the Fail2ban service starting, including the observer and database initialization. The log entries are as follows:

```
[bahis@server.bahis.net ~]$ sudo tail -f /var/log/fail2ban.log
[sudo] password for bahis:
2026-02-13 13:07:14,014 fail2ban.server [50210]: INFO -----
-----
2026-02-13 13:07:14,014 fail2ban.server [50210]: INFO Starting Fail2ban v1.1.0
2026-02-13 13:07:14,014 fail2ban.observer [50210]: INFO Observer start...
2026-02-13 13:07:14,029 fail2ban.database [50210]: INFO Connected to fail2ban persis
nt database '/var/lib/fail2ban/fail2ban.sqlite3'
2026-02-13 13:07:14,031 fail2ban.database [50210]: WARNING New database created. Version
'4'
```

Рисунок 3: Просмотр журнала fail2ban.log в реальном времени

2.1.3 Контроль журнала событий Fail2ban

- Выполнен `tail -f /var/log/fail2ban.log`
- Зафиксирован запуск Fail2ban 1.1.0
- Инициализирован observer
- Создана база `fail2ban.sqlite3`



The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal displays the output of the command `sudo tail -f /var/log/fail2ban.log`. The log entries show the following sequence of events:

```
[bahis@server.bahis.net ~]$ sudo tail -f /var/log/fail2ban.log
[sudo] password for bahis:
2026-02-13 13:07:14,014 fail2ban.server [50210]: INFO -----
-----
2026-02-13 13:07:14,014 fail2ban.server [50210]: INFO Starting Fail2ban v1.1.0
2026-02-13 13:07:14,014 fail2ban.observer [50210]: INFO Observer start...
2026-02-13 13:07:14,029 fail2ban.database [50210]: INFO Connected to fail2ban persis
nt database '/var/lib/fail2ban/fail2ban.sqlite3'
2026-02-13 13:07:14,031 fail2ban.database [50210]: WARNING New database created. Version
'4'
```

Рисунок 3: Просмотр журнала fail2ban.log в реальном времени

2.1.4 Создание файла локальной конфигурации

- Создан файл `/etc/fail2ban/jail.d/customisation.local`

```
[root@server.bahis.net ~]# touch /etc/fail2ban/jail.d/customisation.local
[root@server.bahis.net ~]# nano /etc/fail2ban/jail.d/customisation.local
[root@server.bahis.net ~]# systemctl restart fail2ban
[root@server.bahis.net ~]# :
```

Рисунок 4: Создание и перезапуск службы после настройки

2.1.4 Создание файла локальной конфигурации

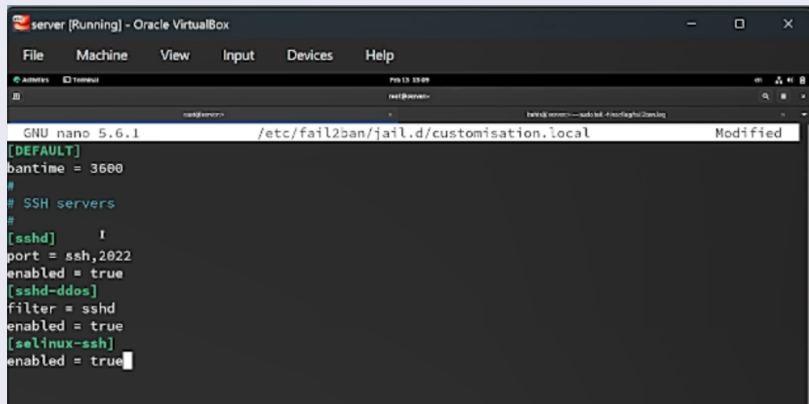
- Создан файл `/etc/fail2ban/jail.d/customisation.local`
- Выполнено редактирование в `nano`

```
[root@server.bahis.net ~]# touch /etc/fail2ban/jail.d/customisation.local
[root@server.bahis.net ~]# nano /etc/fail2ban/jail.d/customisation.local
[root@server.bahis.net ~]# systemctl restart fail2ban
[root@server.bahis.net ~]# :
```

Рисунок 4: Создание и перезапуск службы после настройки

2.1.5 Настройка параметров блокирования и защиты SSH

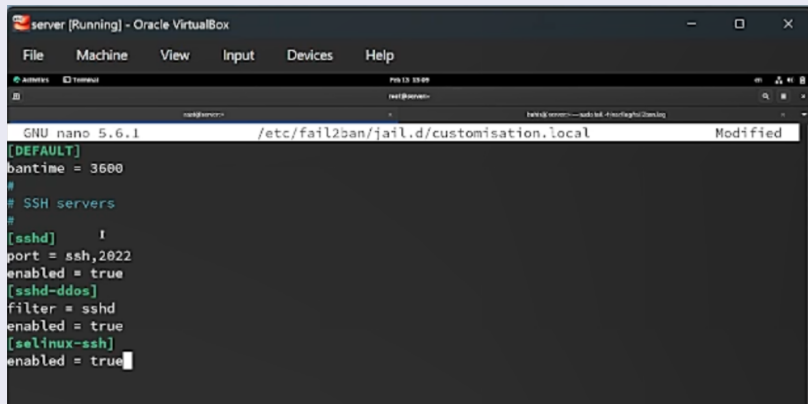
- В [DEFAULT] установлен `bantime = 3600`



```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal
root@server:~#
GNU nano 5.6.1 /etc/fail2ban/jail.d/customisation.local Modified
[DEFAULT]
bantime = 3600
#
# SSH servers
#
[sshd]
port = ssh,2022
enabled = true
[sshd-ddos]
filter = sshd
enabled = true
[selinux-ssh]
enabled = true
```

2.1.5 Настройка параметров блокирования и защиты SSH

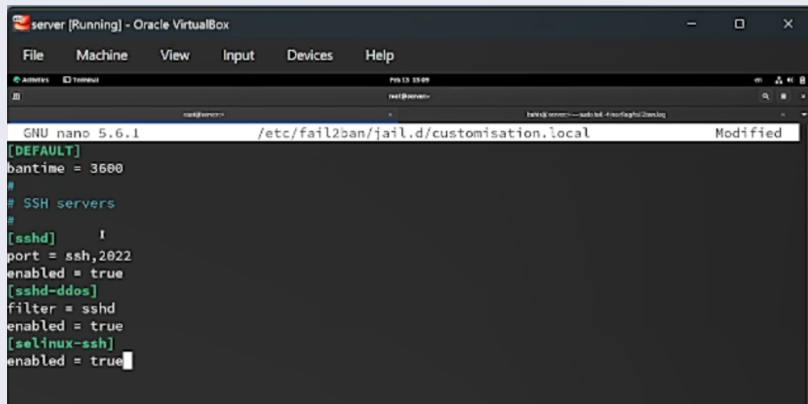
- В [DEFAULT] установлен `bantime = 3600`
- Активированы `[sshd]`, `[sshd-ddos]`, `[selinux-ssh]`



```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal
root@server:~#
GNU nano 5.6.1 /etc/fail2ban/jail.d/customisation.local Modified
[DEFAULT]
bantime = 3600
#
# SSH servers
#
[sshd]
port = ssh,2022
enabled = true
[sshd-ddos]
filter = sshd
enabled = true
[selinux-ssh]
enabled = true
```

2.1.5 Настройка параметров блокирования и защиты SSH

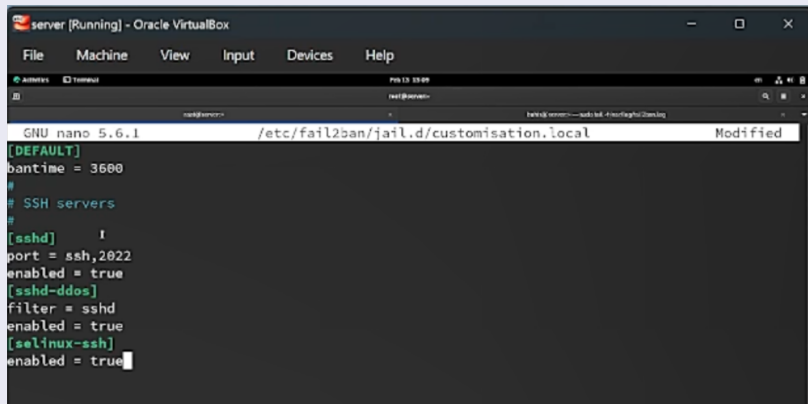
- В [DEFAULT] установлен `bantime = 3600`
- Активированы `[sshd]`, `[sshd-ddos]`, `[selinux-ssh]`
- Указан порт `ssh, 2022`



```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal
Fri 13 03:00
root@server:~#
GNU nano 5.6.1 /etc/fail2ban/jail.d/customisation.local Modified
[DEFAULT]
bantime = 3600
#
# SSH servers
#
[sshd]
port = ssh,2022
enabled = true
[sshd-ddos]
filter = sshd
enabled = true
[selinux-ssh]
enabled = true
```

2.1.5 Настройка параметров блокирования и защиты SSH

- В [DEFAULT] установлен `bantime = 3600`
- Активированы `[sshd]`, `[sshd-ddos]`, `[selinux-ssh]`
- Указан порт `ssh, 2022`
- Для всех секций `enabled = true`



```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal
Fri 13 03:00
root@server:~#
GNU nano 5.6.1 /etc/fail2ban/jail.d/customisation.local Modified
[DEFAULT]
bantime = 3600
#
# SSH servers
#
[sshd]
port = ssh,2022
enabled = true
[sshd-ddos]
filter = sshd
enabled = true
[selinux-ssh]
enabled = true
```

2.1.6 Перезапуск службы Fail2ban

- Выполнен `systemctl restart fail2ban`

```
[root@server.bahis.net ~]# touch /etc/fail2ban/jail.d/customisation.local
[root@server.bahis.net ~]# nano /etc/fail2ban/jail.d/customisation.local
[root@server.bahis.net ~]# systemctl restart fail2ban
[root@server.bahis.net ~]# :
```

Рисунок 6: Создание и перезапуск службы после настройки

2.1.6 Перезапуск службы Fail2ban

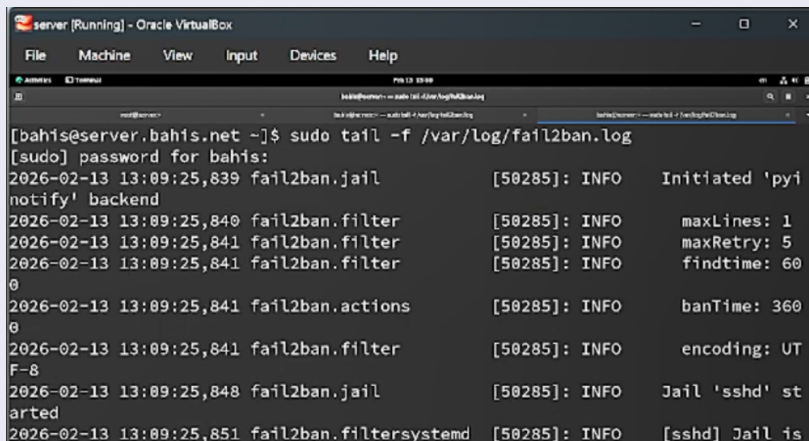
- Выполнен `systemctl restart fail2ban`
- Применена новая конфигурация

```
[root@server.bahis.net ~]# touch /etc/fail2ban/jail.d/customisation.local
[root@server.bahis.net ~]# nano /etc/fail2ban/jail.d/customisation.local
[root@server.bahis.net ~]# systemctl restart fail2ban
[root@server.bahis.net ~]#
```

Рисунок 6: Создание и перезапуск службы после настройки

2.1.7 Просмотр журнала после настройки SSH

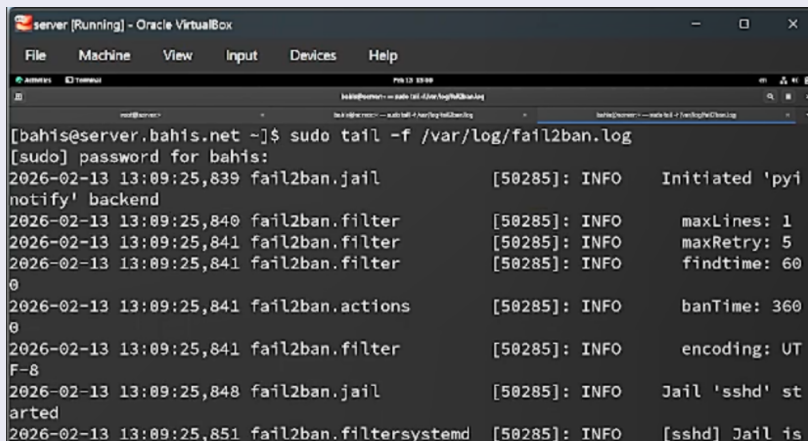
- Инициализирован backend pyinotify



```
server [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
bahis@server: ~$ sudo tail -f /var/log/fail2ban.log
[sudo] password for bahis:
2026-02-13 13:09:25,839 fail2ban.jail [50285]: INFO Initiated 'pyinotify' backend
2026-02-13 13:09:25,840 fail2ban.filter [50285]: INFO maxLines: 1
2026-02-13 13:09:25,841 fail2ban.filter [50285]: INFO maxRetry: 5
2026-02-13 13:09:25,841 fail2ban.filter [50285]: INFO findtime: 60
2026-02-13 13:09:25,841 fail2ban.actions [50285]: INFO banTime: 360
2026-02-13 13:09:25,841 fail2ban.filter [50285]: INFO encoding: UTF-8
2026-02-13 13:09:25,848 fail2ban.jail [50285]: INFO Jail 'sshd' started
2026-02-13 13:09:25,851 fail2ban.filtersystemd [50285]: INFO [sshd] Jail is
```

2.1.7 Просмотр журнала после настройки SSH

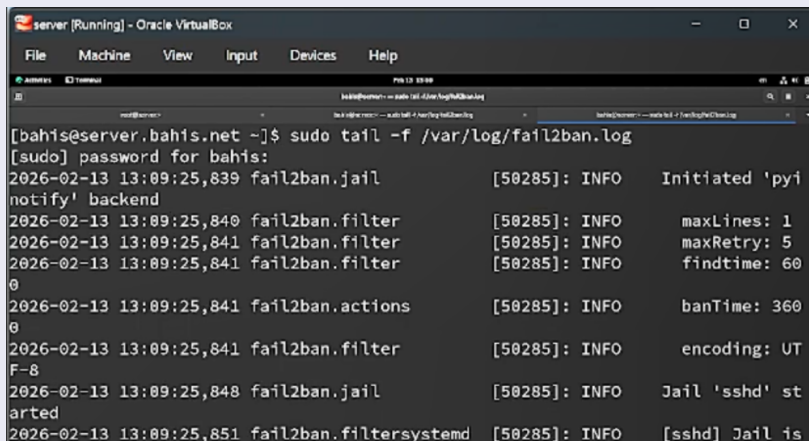
- Инициализирован backend pyinotify
- Установлены параметры maxRetry, findtime, banTime = 3600



```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
bahis@server: ~$ sudo tail -f /var/log/fail2ban.log
[sudo] password for bahis:
2026-02-13 13:09:25,839 fail2ban.jail [50285]: INFO Initiated 'pyi
notify' backend
2026-02-13 13:09:25,840 fail2ban.filter [50285]: INFO maxLines: 1
2026-02-13 13:09:25,841 fail2ban.filter [50285]: INFO maxRetry: 5
2026-02-13 13:09:25,841 fail2ban.filter [50285]: INFO findtime: 60
0
2026-02-13 13:09:25,841 fail2ban.actions [50285]: INFO banTime: 360
0
2026-02-13 13:09:25,841 fail2ban.filter [50285]: INFO encoding: UT
F-8
2026-02-13 13:09:25,848 fail2ban.jail [50285]: INFO Jail 'sshd' st
arted
2026-02-13 13:09:25,851 fail2ban.filtersystemd [50285]: INFO [sshd] Jail is
```

2.1.7 Просмотр журнала после настройки SSH

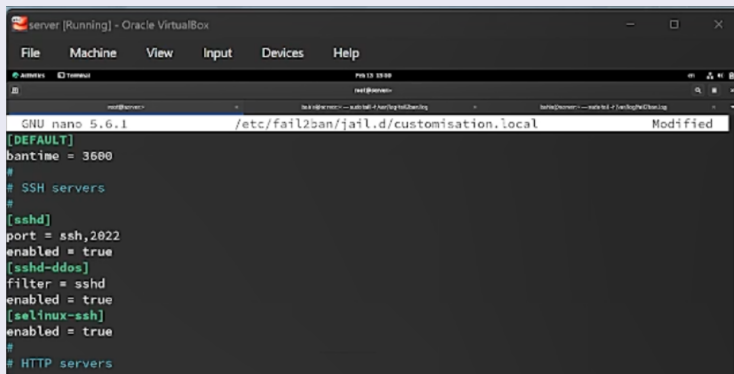
- Инициализирован backend pyinotify
- Установлены параметры maxRetry, findtime, banTime = 3600
- Запущены jail-модули sshd, selinux-ssh, sshd-ddos



```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
bahis@server:~$ sudo tail -f /var/log/fail2ban.log
[sudo] password for bahis:
2026-02-13 13:09:25,839 fail2ban.jail [50285]: INFO Initiated 'pyinotify' backend
2026-02-13 13:09:25,840 fail2ban.filter [50285]: INFO maxLines: 1
2026-02-13 13:09:25,841 fail2ban.filter [50285]: INFO maxRetry: 5
2026-02-13 13:09:25,841 fail2ban.filter [50285]: INFO findtime: 60
2026-02-13 13:09:25,841 fail2ban.actions [50285]: INFO banTime: 360
2026-02-13 13:09:25,841 fail2ban.filter [50285]: INFO encoding: UTF-8
2026-02-13 13:09:25,848 fail2ban.jail [50285]: INFO Jail 'sshd' started
2026-02-13 13:09:25,851 fail2ban.filtersystemd [50285]: INFO [sshd] Jail is
```

2.1.8 Включение защиты HTTP

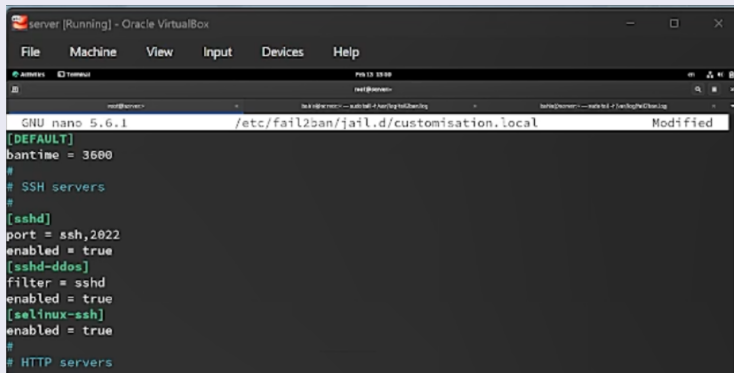
- Активированы jail-модули: apache-auth, apache-badbots, apache-noscript, apache-overflows, apache-nohome, apache-botsearch, apache-fakegooglebot, apache-modsecurity, apache-shellshock



```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Feb 12 15:00
root@server~#
GNU nano 5.6.1 /etc/fail2ban/jail.d/customisation.local Modified
[DEFAULT]
bantime = 3600
#
# SSH servers
#
[sshd]
port = ssh,2022
enabled = true
[sshd-ddos]
filter = sshd
enabled = true
[selinux-ssh]
enabled = true
#
# HTTP servers
```

2.1.8 Включение защиты HTTP

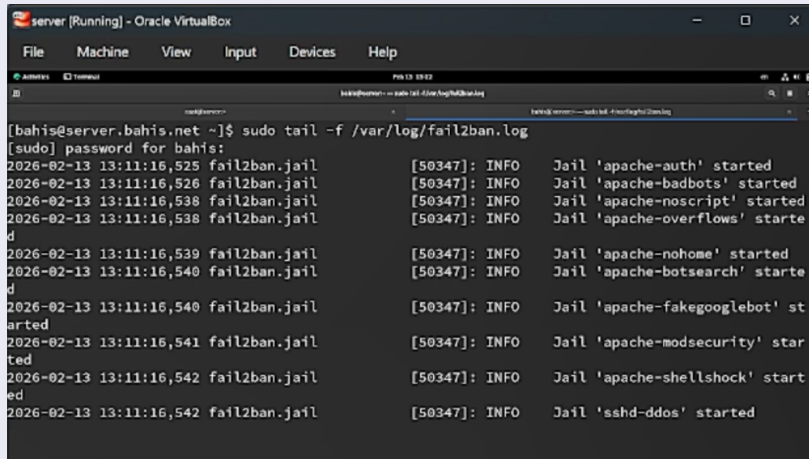
- Активированы jail-модули: apache-auth, apache-badbots, apache-noscript, apache-overflows, apache-nohome, apache-botsearch, apache-fakegooglebot, apache-modsecurity, apache-shellshock
- Для всех `enabled = true`



```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Feb 12 15:00
root@server~#
GNU nano 5.6.1 /etc/fail2ban/jail.d/customisation.local Modified
[DEFAULT]
bantime = 3600
#
# SSH servers
#
[sshd]
port = ssh,2022
enabled = true
[sshd-ddos]
filter = sshd
enabled = true
[selinux-ssh]
enabled = true
#
# HTTP servers
```

2.1.9 Перезапуск после настройки HTTP

- Выполнен `systemctl restart fail2ban`

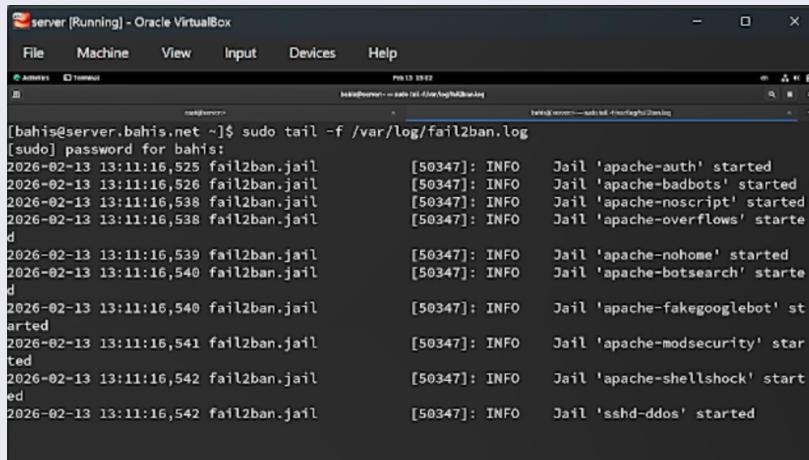


The screenshot shows a terminal window titled "server [Running] - Oracle VM VirtualBox". The terminal output displays the command `sudo tail -f /var/log/fail2ban.log` and its results. The output shows several log entries from `fail2ban.jail` indicating the start of various jails. The jails listed are: `apache-auth`, `apache-badbots`, `apache-noscript`, `apache-overflows`, `apache-nohome`, `apache-botsearch`, `apache-fakegooglebot`, `apache-modsecurity`, `apache-shellshock`, and `sshd-ddos`. Each entry is preceded by a timestamp and the source `fail2ban.jail`.

```
[bahis@server.bahis.net ~]$ sudo tail -f /var/log/fail2ban.log
[sudo] password for bahis:
2026-02-13 13:11:16,525 fail2ban.jail [50347]: INFO Jail 'apache-auth' started
2026-02-13 13:11:16,526 fail2ban.jail [50347]: INFO Jail 'apache-badbots' started
2026-02-13 13:11:16,538 fail2ban.jail [50347]: INFO Jail 'apache-noscript' started
2026-02-13 13:11:16,538 fail2ban.jail [50347]: INFO Jail 'apache-overflows' starte
d
2026-02-13 13:11:16,539 fail2ban.jail [50347]: INFO Jail 'apache-nohome' started
2026-02-13 13:11:16,540 fail2ban.jail [50347]: INFO Jail 'apache-botsearch' starte
d
2026-02-13 13:11:16,540 fail2ban.jail [50347]: INFO Jail 'apache-fakegooglebot' st
arted
2026-02-13 13:11:16,541 fail2ban.jail [50347]: INFO Jail 'apache-modsecurity' star
ted
2026-02-13 13:11:16,542 fail2ban.jail [50347]: INFO Jail 'apache-shellshock' start
ed
2026-02-13 13:11:16,542 fail2ban.jail [50347]: INFO Jail 'sshd-ddos' started
```

2.1.9 Перезапуск после настройки HTTP

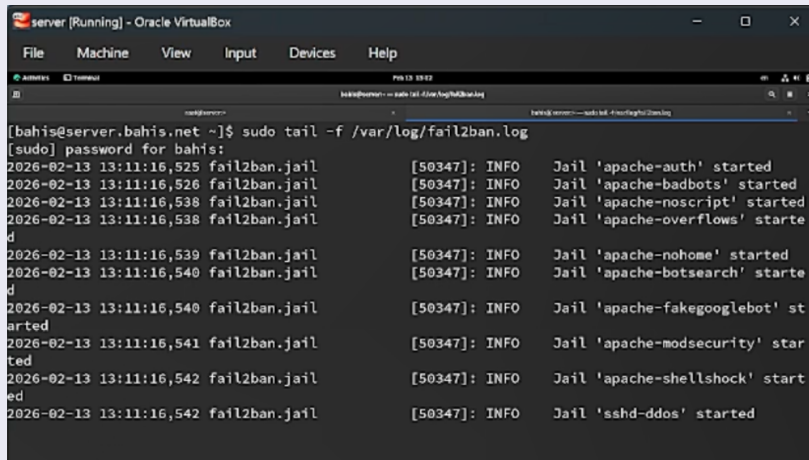
- Выполнен `systemctl restart fail2ban`
- Загружены HTTP jail-модули



```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Feb 13 13:02
bahis@server ~$ sudo tail -f /var/log/fail2ban.log
[sudo] password for bahis:
2026-02-13 13:11:16,525 fail2ban.jail [50347]: INFO Jail 'apache-auth' started
2026-02-13 13:11:16,526 fail2ban.jail [50347]: INFO Jail 'apache-badbots' started
2026-02-13 13:11:16,538 fail2ban.jail [50347]: INFO Jail 'apache-noscript' started
2026-02-13 13:11:16,538 fail2ban.jail [50347]: INFO Jail 'apache-overflows' starte
d
2026-02-13 13:11:16,539 fail2ban.jail [50347]: INFO Jail 'apache-nohome' started
2026-02-13 13:11:16,540 fail2ban.jail [50347]: INFO Jail 'apache-botsearch' starte
d
2026-02-13 13:11:16,540 fail2ban.jail [50347]: INFO Jail 'apache-fakegooglebot' st
arted
2026-02-13 13:11:16,541 fail2ban.jail [50347]: INFO Jail 'apache-modsecurity' star
ted
2026-02-13 13:11:16,542 fail2ban.jail [50347]: INFO Jail 'apache-shellshock' start
ed
2026-02-13 13:11:16,542 fail2ban.jail [50347]: INFO Jail 'sshd-ddos' started
```

2.1.10 Контроль журнала HTTP-защиты

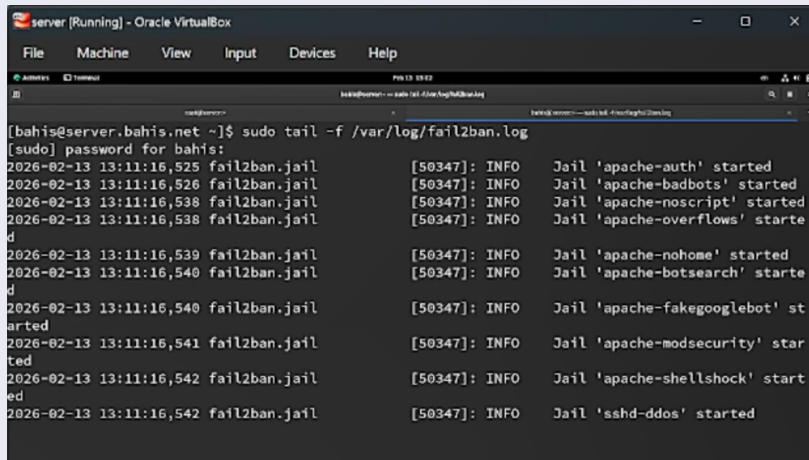
- В журнале зафиксирован запуск всех apache-* jail



```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Feb 13 13:02
bahis@server: ~$ sudo tail -f /var/log/fail2ban.log
[sudo] password for bahis:
2026-02-13 13:11:16,525 fail2ban.jail [50347]: INFO Jail 'apache-auth' started
2026-02-13 13:11:16,526 fail2ban.jail [50347]: INFO Jail 'apache-badbots' started
2026-02-13 13:11:16,538 fail2ban.jail [50347]: INFO Jail 'apache-noscript' started
2026-02-13 13:11:16,538 fail2ban.jail [50347]: INFO Jail 'apache-overflows' starte
d
2026-02-13 13:11:16,539 fail2ban.jail [50347]: INFO Jail 'apache-nohome' started
2026-02-13 13:11:16,540 fail2ban.jail [50347]: INFO Jail 'apache-botsearch' starte
d
2026-02-13 13:11:16,540 fail2ban.jail [50347]: INFO Jail 'apache-fakegooglebot' st
arted
2026-02-13 13:11:16,541 fail2ban.jail [50347]: INFO Jail 'apache-modsecurity' star
ted
2026-02-13 13:11:16,542 fail2ban.jail [50347]: INFO Jail 'apache-shellshock' start
ed
2026-02-13 13:11:16,542 fail2ban.jail [50347]: INFO Jail 'sshd-ddos' started
```


2.1.10 Контроль журнала HTTP-защиты

- В журнале зафиксирован запуск всех apache-* jail
- Подтверждена активация sshd-ddos



```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Feb 13 13:02
bahis@server ~$ sudo tail -f /var/log/fail2ban.log
[sudo] password for bahis:
2026-02-13 13:11:16,525 fail2ban.jail [50347]: INFO Jail 'apache-auth' started
2026-02-13 13:11:16,526 fail2ban.jail [50347]: INFO Jail 'apache-badbots' started
2026-02-13 13:11:16,538 fail2ban.jail [50347]: INFO Jail 'apache-noscript' started
2026-02-13 13:11:16,538 fail2ban.jail [50347]: INFO Jail 'apache-overflows' starte
d
2026-02-13 13:11:16,539 fail2ban.jail [50347]: INFO Jail 'apache-nohome' started
2026-02-13 13:11:16,540 fail2ban.jail [50347]: INFO Jail 'apache-botsearch' starte
d
2026-02-13 13:11:16,540 fail2ban.jail [50347]: INFO Jail 'apache-fakegooglebot' st
arted
2026-02-13 13:11:16,541 fail2ban.jail [50347]: INFO Jail 'apache-modsecurity' star
ted
2026-02-13 13:11:16,542 fail2ban.jail [50347]: INFO Jail 'apache-shellshock' start
ed
2026-02-13 13:11:16,542 fail2ban.jail [50347]: INFO Jail 'sshd-ddos' started
```

2.1.11 Включение защиты почтовых сервисов

- Активированы jail-модули: postfix, postfix-rbl, dovecot, postfix-sasl

```
#  
# Mail servers  
#  
[postfix]  
enabled = true  
[postfix-rbl]  
enabled = true  
[dovecot]  
enabled = true  
[postfix-sasl]  
enabled = true
```

Рисунок 11: Конфигурация jail-модулей для почтовых сервисов

2.1.11 Включение защиты почтовых сервисов

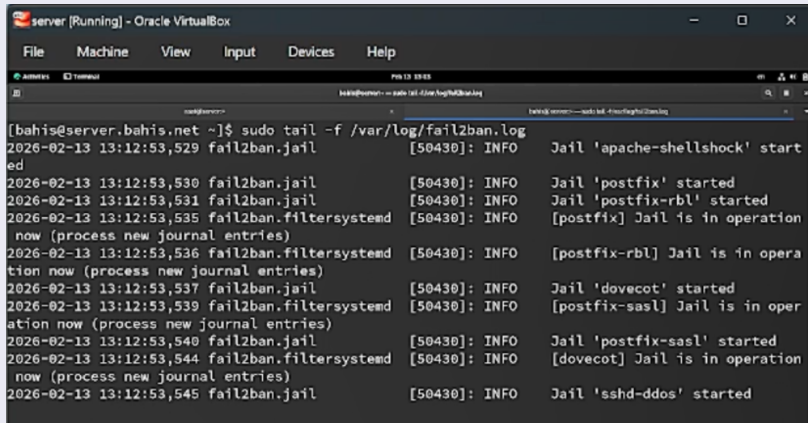
- Активированы jail-модули: postfix, postfix-rbl, dovecot, postfix-sasl
- Установлено `enabled = true`

```
#  
# Mail servers  
#  
[postfix]  
enabled = true  
[postfix-rbl]  
enabled = true  
[dovecot]  
enabled = true  
[postfix-sasl]  
enabled = true
```

Рисунок 11: Конфигурация jail-модулей для почтовых сервисов

2.1.12 Перезапуск после настройки почтовых служб

- Выполнен перезапуск fail2ban



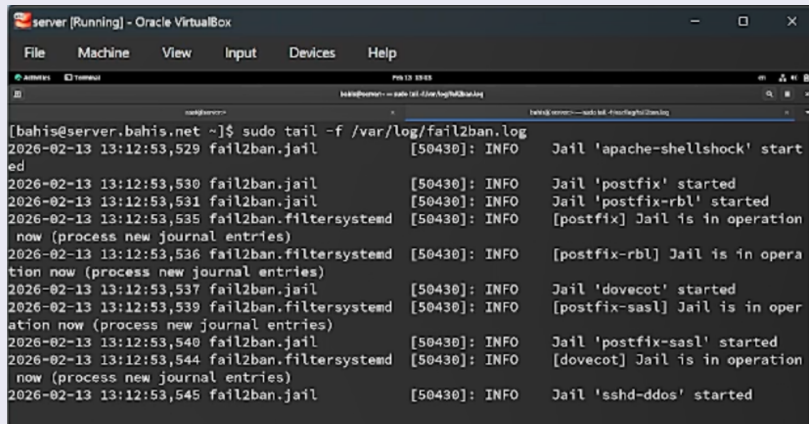
The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The user is logged in as "bahis" on a server named "bahis.net". The command executed is `sudo tail -f /var/log/fail2ban.log`. The log output shows the following events:

```
[bahis@server.bahis.net ~]$ sudo tail -f /var/log/fail2ban.log
2026-02-13 13:12:53,529 fail2ban.jail [50430]: INFO Jail 'apache-shellshock' started
2026-02-13 13:12:53,530 fail2ban.jail [50430]: INFO Jail 'postfix' started
2026-02-13 13:12:53,531 fail2ban.jail [50430]: INFO Jail 'postfix-rbl' started
2026-02-13 13:12:53,535 fail2ban.filtersystemd [50430]: INFO [postfix] Jail is in operation
now (process new journal entries)
2026-02-13 13:12:53,536 fail2ban.filtersystemd [50430]: INFO [postfix-rbl] Jail is in operation
now (process new journal entries)
2026-02-13 13:12:53,537 fail2ban.jail [50430]: INFO Jail 'dovecot' started
2026-02-13 13:12:53,539 fail2ban.filtersystemd [50430]: INFO [postfix-sasl] Jail is in operation
now (process new journal entries)
2026-02-13 13:12:53,540 fail2ban.jail [50430]: INFO Jail 'postfix-sasl' started
2026-02-13 13:12:53,544 fail2ban.filtersystemd [50430]: INFO [dovecot] Jail is in operation
now (process new journal entries)
2026-02-13 13:12:53,545 fail2ban.jail [50430]: INFO Jail 'sshd-ddos' started
```

Рисунок 12: Журнал запуска jail-модулей почтовых сервисов

2.1.12 Перезапуск после настройки почтовых служб

- Выполнен перезапуск fail2ban
- Загружены почтовые jail-модули



```
[bahis@server.bahis.net ~]$ sudo tail -f /var/log/fail2ban.log
2026-02-13 13:12:53,529 fail2ban.jail [50430]: INFO Jail 'apache-shellshock' started
2026-02-13 13:12:53,530 fail2ban.jail [50430]: INFO Jail 'postfix' started
2026-02-13 13:12:53,531 fail2ban.jail [50430]: INFO Jail 'postfix-rbl' started
2026-02-13 13:12:53,535 fail2ban.filtersystemd [50430]: INFO [postfix] Jail is in operation
now (process new journal entries)
2026-02-13 13:12:53,536 fail2ban.filtersystemd [50430]: INFO [postfix-rbl] Jail is in operation
now (process new journal entries)
2026-02-13 13:12:53,537 fail2ban.jail [50430]: INFO Jail 'dovecot' started
2026-02-13 13:12:53,539 fail2ban.filtersystemd [50430]: INFO [postfix-sasl] Jail is in operation
now (process new journal entries)
2026-02-13 13:12:53,540 fail2ban.jail [50430]: INFO Jail 'postfix-sasl' started
2026-02-13 13:12:53,544 fail2ban.filtersystemd [50430]: INFO [dovecot] Jail is in operation
now (process new journal entries)
2026-02-13 13:12:53,545 fail2ban.jail [50430]: INFO Jail 'sshd-ddos' started
```

Рисунок 12: Журнал запуска jail-модулей почтовых сервисов

2.1.13 Контроль журнала почтовой защиты

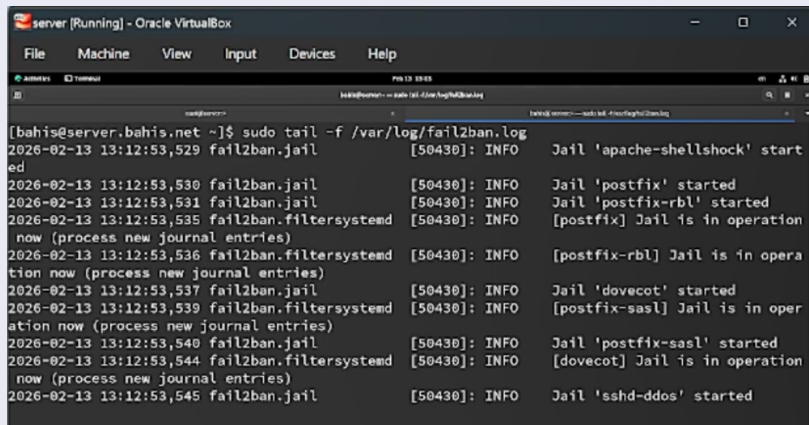
- Зафиксировано Jail is in operation now

```
[bahis@server.bahis.net ~]$ sudo tail -f /var/log/fail2ban.log
2026-02-13 13:12:53,529 fail2ban.jail [50430]: INFO Jail 'apache-shellshock' started
2026-02-13 13:12:53,530 fail2ban.jail [50430]: INFO Jail 'postfix' started
2026-02-13 13:12:53,531 fail2ban.jail [50430]: INFO Jail 'postfix-rbl' started
2026-02-13 13:12:53,535 fail2ban.filtersystemd [50430]: INFO [postfix] Jail is in operation
now (process new journal entries)
2026-02-13 13:12:53,536 fail2ban.filtersystemd [50430]: INFO [postfix-rbl] Jail is in operation
now (process new journal entries)
2026-02-13 13:12:53,537 fail2ban.jail [50430]: INFO Jail 'dovecot' started
2026-02-13 13:12:53,539 fail2ban.filtersystemd [50430]: INFO [postfix-sasl] Jail is in operation
now (process new journal entries)
2026-02-13 13:12:53,540 fail2ban.jail [50430]: INFO Jail 'postfix-sasl' started
2026-02-13 13:12:53,544 fail2ban.filtersystemd [50430]: INFO [dovecot] Jail is in operation
now (process new journal entries)
2026-02-13 13:12:53,545 fail2ban.jail [50430]: INFO Jail 'sshd-ddos' started
```

Рисунок 13: Подтверждение работы jail-модулей почтовых сервисов

2.1.13 Контроль журнала почтовой защиты

- Зафиксировано Jail is in operation now
- Подтверждена активация postfix, dovecot, postfix-sasl



```
[bahis@server.bahis.net ~]$ sudo tail -f /var/log/fail2ban.log
2026-02-13 13:12:53,529 fail2ban.jail [50430]: INFO Jail 'apache-shellshock' started
2026-02-13 13:12:53,530 fail2ban.jail [50430]: INFO Jail 'postfix' started
2026-02-13 13:12:53,531 fail2ban.jail [50430]: INFO Jail 'postfix-rbl' started
2026-02-13 13:12:53,535 fail2ban.filtersystemd [50430]: INFO [postfix] Jail is in operation now (process new journal entries)
2026-02-13 13:12:53,536 fail2ban.filtersystemd [50430]: INFO [postfix-rbl] Jail is in operation now (process new journal entries)
2026-02-13 13:12:53,537 fail2ban.jail [50430]: INFO Jail 'dovecot' started
2026-02-13 13:12:53,539 fail2ban.filtersystemd [50430]: INFO [postfix-sasl] Jail is in operation now (process new journal entries)
2026-02-13 13:12:53,540 fail2ban.jail [50430]: INFO Jail 'postfix-sasl' started
2026-02-13 13:12:53,544 fail2ban.filtersystemd [50430]: INFO [dovecot] Jail is in operation now (process new journal entries)
2026-02-13 13:12:53,545 fail2ban.jail [50430]: INFO Jail 'sshd-ddos' started
```

Рисунок 13: Подтверждение работы jail-модулей почтовых сервисов

2.2 Проверка работы Fail2ban

2.2.1 Просмотр общего статуса

- Выполнен `fail2ban-client status`

```
[root@server.bahis.net ~]# fail2ban-client status
Status
|- Number of jail:      16
|- Jail list:  apache-auth, apache-badbots, apache-botsearch, apache-fakegooglebot, apache-mod
security, apache-nohome, apache-noscript, apache-overflows, apache-shellshock, dovecot, postfix
, postfix-rbl, postfix-sasl, selinux-ssh, sshd, sshd-ddos
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:     0
|   '- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
- Actions
|   |- Currently banned: 0
|   |- Total banned:     0
|   '- Banned IP list:
[root@server.bahis.net ~]# fail2ban-client set sshd maxretry 2
2
[root@server.bahis.net ~]#
```


2.2 Проверка работы Fail2ban

2.2.1 Просмотр общего статуса

- Выполнен `fail2ban-client status`
- Активировано 16 jail-модулей

```
[root@server.bahis.net ~]# fail2ban-client status
Status
|- Number of jail:      16
|- Jail list:  apache-auth, apache-badbots, apache-botsearch, apache-fakegooglebot, apache-mod
security, apache-nohome, apache-noscript, apache-overflows, apache-shellshock, dovecot, postfix
, postfix-rbl, postfix-sasl, selinux-ssh, sshd, sshd-ddos
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:    0
|   '- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
- Actions
|   |- Currently banned: 0
|   |- Total banned:    0
|   '- Banned IP list:
[root@server.bahis.net ~]# fail2ban-client set sshd maxretry 2
2
[root@server.bahis.net ~]#
```

2.2 Проверка работы Fail2ban

2.2.1 Просмотр общего статуса

- Выполнен `fail2ban-client status`
- Активировано 16 jail-модулей
- Отображены `apache-*`, `sshd`, `dovecot`, `postfix`

```
[root@server.bahis.net ~]# fail2ban-client status
Status
|- Number of jail:      16
|- Jail list:  apache-auth, apache-badbots, apache-botsearch, apache-fakegooglebot, apache-mod
security, apache-nohome, apache-noscript, apache-overflows, apache-shellshock, dovecot, postfix
, postfix-rbl, postfix-sasl, selinux-ssh, sshd, sshd-ddos
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:    0
|   '- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
- Actions
|   |- Currently banned: 0
|   |- Total banned:    0
|   '- Banned IP list:
[root@server.bahis.net ~]# fail2ban-client set sshd maxretry 2
2
[root@server.bahis.net ~]#
```

2.2.2 Просмотр статуса SSH

- Выполнен `fail2ban-client status sshd`

```
[root@server.bahis.net ~]# fail2ban-client status
Status
|- Number of jail:      16
|- Jail list:  apache-auth, apache-badbots, apache-botsearch, apache-fakegooglebot, apache-mod
security, apache-nohome, apache-noscript, apache-overflows, apache-shellshock, dovecot, postfix
, postfix-rbl, postfix-sasl, selinux-ssh, sshd, sshd-ddos
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:     0
|   '- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
- Actions
|   |- Currently banned: 0
|   |- Total banned:     0
|   '- Banned IP list:
[root@server.bahis.net ~]# fail2ban-client set sshd maxretry 2
2
[root@server.bahis.net ~]#
```

Рисунок 15: Статус jail-модуля sshd до проверки

2.2.2 Просмотр статуса SSH

- Выполнен `fail2ban-client status sshd`
- Currently banned: 0

```
[root@server.bahis.net ~]# fail2ban-client status
Status
|- Number of jail:      16
|- Jail list:  apache-auth, apache-badbots, apache-botsearch, apache-fakegooglebot, apache-mod
security, apache-nohome, apache-noscript, apache-overflows, apache-shellshock, dovecot, postfix
, postfix-rbl, postfix-sasl, selinux-ssh, sshd, sshd-ddos
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:     0
|   '- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
- Actions
|   |- Currently banned: 0
|   |- Total banned:     0
|   '- Banned IP list:
[root@server.bahis.net ~]# fail2ban-client set sshd maxretry 2
2
[root@server.bahis.net ~]#
```

Рисунок 15: Статус jail-модуля sshd до проверки

2.2.2 Просмотр статуса SSH

- Выполнен `fail2ban-client status sshd`
- Currently banned: 0
- Total banned: 0

```
[root@server.bahis.net ~]# fail2ban-client status
Status
|- Number of jail:      16
|- Jail list:  apache-auth, apache-badbots, apache-botsearch, apache-fakegooglebot, apache-mod
security, apache-nohome, apache-noscript, apache-overflows, apache-shellshock, dovecot, postfix
, postfix-rbl, postfix-sasl, selinux-ssh, sshd, sshd-ddos
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:     0
|   '- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
- Actions
|   |- Currently banned: 0
|   |- Total banned:     0
|   '- Banned IP list:
[root@server.bahis.net ~]# fail2ban-client set sshd maxretry 2
2
[root@server.bahis.net ~]#
```

Рисунок 15: Статус jail-модуля sshd до проверки

2.2.3 Установка maxretry

- Выполнено `fail2ban-client set sshd maxretry 2`

```
[root@server.bahis.net ~]# fail2ban-client status
Status
|- Number of jail:      16
|- Jail list:  apache-auth, apache-badbots, apache-botsearch, apache-fakegooglebot, apache-mod
security, apache-nohome, apache-noscript, apache-overflows, apache-shellshock, dovecot, postfix
, postfix-rbl, postfix-sasl, selinux-ssh, sshd, sshd-ddos
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:    0
|  '- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
- Actions
|  |- Currently banned: 0
|  |- Total banned:    0
|  '- Banned IP list:
[root@server.bahis.net ~]# fail2ban-client set sshd maxretry 2
2
[root@server.bahis.net ~]#
```

Рисунок 16: Установка параметра maxretry для sshd

2.2.3 Установка maxretry

- Выполнено `fail2ban-client set sshd maxretry 2`
- После 2 ошибок IP должен блокироваться

```
[root@server.bahis.net ~]# fail2ban-client status
Status
|- Number of jail:      16
|- Jail list:  apache-auth, apache-badbots, apache-botsearch, apache-fakegooglebot, apache-mod
security, apache-nohome, apache-noscript, apache-overflows, apache-shellshock, dovecot, postfix
, postfix-rbl, postfix-sasl, selinux-ssh, sshd, sshd-ddos
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:     0
|   '- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
- Actions
|   |- Currently banned: 0
|   |- Total banned:     0
|   '- Banned IP list:
[root@server.bahis.net ~]# fail2ban-client set sshd maxretry 2
2
[root@server.bahis.net ~]#
```

Рисунок 16: Установка параметра maxretry для sshd

2.2.4 Попытка входа с неверным паролем

2.2.5 Проверка блокировки IP

- Выполнен `fail2ban-client status sshd`

```
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed: 0
|   '- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
- Actions
    |- Currently banned: 0
    |- Total banned: 0
    '- Banned IP list:
[root@server.bahis.net ~]# fail2ban-client set sshd maxretry 2
```

Рисунок 17: Блокировка IP-адреса клиента в sshd

2.2.5 Проверка блокировки IP

- Выполнен `fail2ban-client status sshd`
- Currently banned: 1

```
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed: 0
|   '- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
- Actions
    |- Currently banned: 0
    |- Total banned: 0
    '- Banned IP list:
[root@server.bahis.net ~]# fail2ban-client set sshd maxretry 2
```

Рисунок 17: Блокировка IP-адреса клиента в sshd

2.2.5 Проверка блокировки IP

- Выполнен `fail2ban-client status sshd`
- Currently banned: 1
- IP 192.168.1.30 в списке блокировки

```
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed: 0
|   '- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
- Actions
    |- Currently banned: 0
    |- Total banned: 0
    '- Banned IP list:
[root@server.bahis.net ~]# fail2ban-client set sshd maxretry 2
```

Рисунок 17: Блокировка IP-адреса клиента в sshd

2.2.6 Разблокировка IP

- Выполнено `fail2ban-client set sshd unbanip 192.168.1.30`

```
[root@server.bahis.net ~]# fail2ban-client set sshd unbanip 192.168.1.30
θ
```

Рисунок 18: Команда разблокировки IP-адреса клиента

2.2.7 Проверка снятия блокировки

- Повторный status sshd

```
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed: 0
| '- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
- Actions
  |- Currently banned: 0
  |- Total banned: 0
  '- Banned IP list:
[root@server.bahis.net ~]#
```

Рисунок 19: Статус sshd после снятия блокировки

2.2.7 Проверка снятия блокировки

- Повторный status sshd
- Currently banned: 0

```
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed: 0
| '- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
- Actions
  |- Currently banned: 0
  |- Total banned: 0
  '- Banned IP list:
[root@server.bahis.net ~]#
```

Рисунок 19: Статус sshd после снятия блокировки

2.2.7 Проверка снятия блокировки

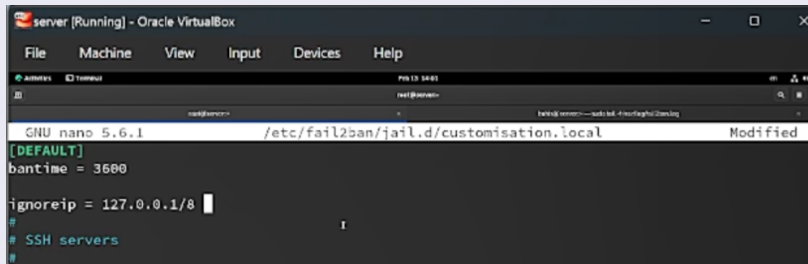
- Повторный status sshd
- Currently banned: 0
- Список пуст

```
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:    0
|   '- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
- Actions
    |- Currently banned: 0
    |- Total banned:    0
    '- Banned IP list:
[root@server.bahis.net ~]#
```

Рисунок 19: Статус sshd после снятия блокировки

2.2.8 Добавление ignoreip

- В [DEFAULT] добавлен ignoreip = 127.0.0.1/8 192.168.1.30



The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal is running the nano text editor, editing the file `/etc/fail2ban/jail.d/customisation.local`. The file is in the `[DEFAULT]` section. The current content of the file is:

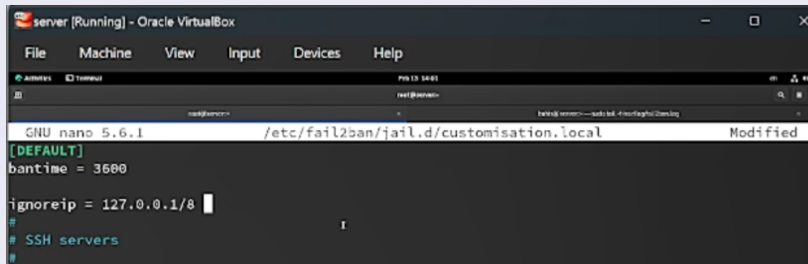
```
GNU nano 5.6.1 /etc/fail2ban/jail.d/customisation.local Modified
[DEFAULT]
bantime = 3600

ignoreip = 127.0.0.1/8
#
# SSH servers
#
```

Рисунок 20: Добавление параметра ignoreip в customisation.local

2.2.8 Добавление ignoreip

- В [DEFAULT] добавлен ignoreip = 127.0.0.1/8 192.168.1.30
- IP клиента исключён из блокировки



The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal is running the nano text editor to edit the file `/etc/fail2ban/jail.d/customisation.local`. The file is in the `[DEFAULT]` section. The current content of the file is:

```
GNU nano 5.6.1 /etc/fail2ban/jail.d/customisation.local Modified
bantime = 3600

ignoreip = 127.0.0.1/8
#
# SSH servers
#
```

Рисунок 20: Добавление параметра ignoreip в customisation.local

2.2.9 Просмотр журнала ignoreip

- В журнале отображено Ignore 192.168.1.30 by ip

```
[bahis@server.bahis.net ~]$ sudo tail -f /var/log/fail2ban.log
[sudo] password for bahis:
2026-02-13 14:02:40,110 fail2ban.jail [50545]: INFO Jail 'apache-shellshock' started
2026-02-13 14:02:40,110 fail2ban.filtersystemd [50545]: INFO [postfix] Jail is in operation
now (process new journal entries)
2026-02-13 14:02:40,111 fail2ban.jail [50545]: INFO Jail 'postfix' started
2026-02-13 14:02:40,111 fail2ban.filtersystemd [50545]: INFO [postfix-rbl] Jail is in operation
now (process new journal entries)
2026-02-13 14:02:40,111 fail2ban.jail [50545]: INFO Jail 'postfix-rbl' started
2026-02-13 14:02:40,112 fail2ban.filtersystemd [50545]: INFO [dovecot] Jail is in operation
now (process new journal entries)
2026-02-13 14:02:40,112 fail2ban.jail [50545]: INFO Jail 'dovecot' started
2026-02-13 14:02:40,112 fail2ban.filtersystemd [50545]: INFO [postfix-sasl] Jail is in operation
now (process new journal entries)
2026-02-13 14:02:40,112 fail2ban.jail [50545]: INFO Jail 'postfix-sasl' started
2026-02-13 14:02:40,113 fail2ban.jail [50545]: INFO Jail 'sshd-ddos' started
```

Рисунок 21: Сообщения журнала о игнорировании IP-адреса

2.2.9 Просмотр журнала ignoreip

- В журнале отображено Ignore 192.168.1.30 by ip
- IP корректно исключён

```
[bahis@server.bahis.net ~]$ sudo tail -f /var/log/fail2ban.log
[sudo] password for bahis:
2026-02-13 14:02:40,110 fail2ban.jail [50545]: INFO Jail 'apache-shellshock' started
2026-02-13 14:02:40,110 fail2ban.filterssystemd [50545]: INFO [postfix] Jail is in operation now (process new journal entries)
2026-02-13 14:02:40,111 fail2ban.jail [50545]: INFO Jail 'postfix' started
2026-02-13 14:02:40,111 fail2ban.filterssystemd [50545]: INFO [postfix-rbl] Jail is in operation now (process new journal entries)
2026-02-13 14:02:40,111 fail2ban.jail [50545]: INFO Jail 'postfix-rbl' started
2026-02-13 14:02:40,112 fail2ban.filterssystemd [50545]: INFO [dovecot] Jail is in operation now (process new journal entries)
2026-02-13 14:02:40,112 fail2ban.jail [50545]: INFO Jail 'dovecot' started
2026-02-13 14:02:40,112 fail2ban.filterssystemd [50545]: INFO [postfix-sasl] Jail is in operation now (process new journal entries)
2026-02-13 14:02:40,112 fail2ban.jail [50545]: INFO Jail 'postfix-sasl' started
2026-02-13 14:02:40,113 fail2ban.jail [50545]: INFO Jail 'sshd-ddos' started
```

Рисунок 21: Сообщения журнала о игнорировании IP-адреса

2.2.10 Повторная проверка SSH

- Повторные ошибки входа не приводят к блокировке

```
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:    0
|  '- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
'- Actions
|  |- Currently banned: 0
|  |- Total banned:    0
|  '- Banned IP list:
```

Рисунок 22: Статус sshd после включения ignoreip

2.2.10 Повторная проверка SSH

- Повторные ошибки входа не приводят к блокировке
- Currently banned: 0

```
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed:    0
| '- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
   |- Currently banned: 0
   |- Total banned:    0
   '- Banned IP list:
```

Рисунок 22: Статус sshd после включения ignoreip

2.2.10 Повторная проверка SSH

- Повторные ошибки входа не приводят к блокировке
- Currently banned: 0
- Список IP пуст

```
[root@server.bahis.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed: 0
| '- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
'- Actions
  |- Currently banned: 0
  |- Total banned: 0
  '- Banned IP list:
```

Рисунок 22: Статус sshd после включения ignoreip

2.3 Внесение изменений в настройки внутреннего

2.3.1 Подготовка конфигурации

- Создан каталог `protect/etc/fail2ban/jail.d`

```
[root@server.bahis.net ~]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/protect/etc/fail2ban/jail.d
[root@server.bahis.net server]# cp -R /etc/fail2ban/jail.d/customisation.local
cp: missing destination file operand after '/etc/fail2ban/jail.d/customisation.local'
Try 'cp --help' for more information.
[root@server.bahis.net server]# cp -R /etc/fail2ban/jail.d/customisation.local /vagrant/provisi
on/server/protect/etc/fail2ban/jail.d/
[root@server.bahis.net server]# cd /vagrant/provision/server
[root@server.bahis.net server]# touch protect.sh
[root@server.bahis.net server]# chmod +x protect.sh
[root@server.bahis.net server]#
```

Рисунок 23: Создание каталога `protect` и копирование `customisation.local`

2.3 Внесение изменений в настройки внутреннего

2.3.1 Подготовка конфигурации

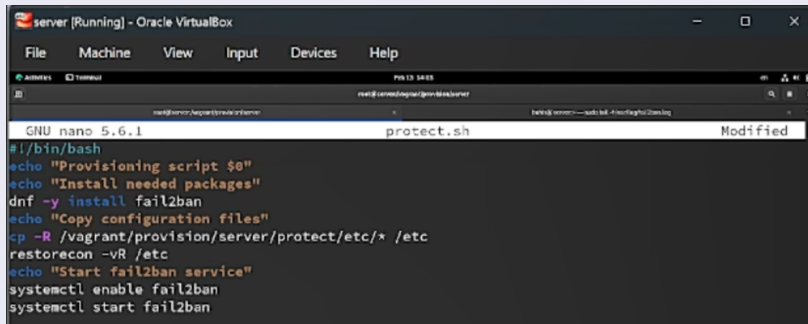
- Создан каталог `protect/etc/fail2ban/jail.d`
- Скопирован `customisation.local`

```
[root@server.bahis.net ~]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/protect/etc/fail2ban/jail.d
[root@server.bahis.net server]# cp -R /etc/fail2ban/jail.d/customisation.local
cp: missing destination file operand after '/etc/fail2ban/jail.d/customisation.local'
Try 'cp --help' for more information.
[root@server.bahis.net server]# cp -R /etc/fail2ban/jail.d/customisation.local /vagrant/provisi
on/server/protect/etc/fail2ban/jail.d/
[root@server.bahis.net server]# cd /vagrant/provision/server
[root@server.bahis.net server]# touch protect.sh
[root@server.bahis.net server]# chmod +x protect.sh
[root@server.bahis.net server]#
```

Рисунок 23: Создание каталога `protect` и копирование `customisation.local`

2.3.2 Создание скрипта protect.sh

- Установка fail2ban



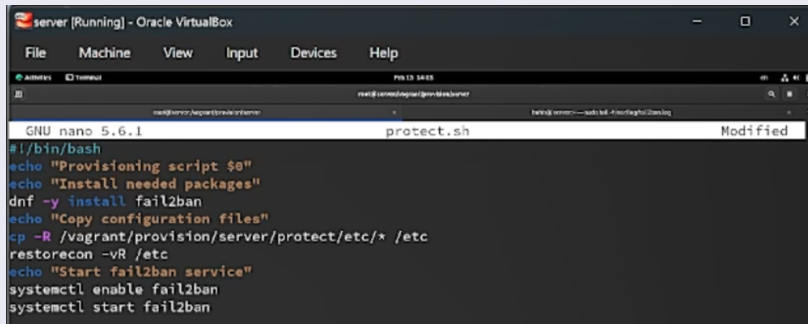
The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal is running the GNU nano 5.6.1 editor, editing a file named "protect.sh". The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install fail2ban
echo "Copy configuration files"
cp -R /vagrant/provision/server/protect/etc/* /etc
restorecon -vR /etc
echo "Start fail2ban service"
systemctl enable fail2ban
systemctl start fail2ban
```

Рисунок 24: Содержимое скрипта protect.sh

2.3.2 Создание скрипта protect.sh

- Установка fail2ban
- Копирование конфигурации в /etc



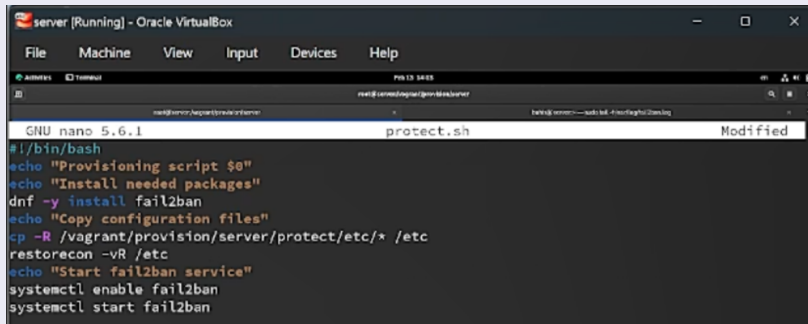
The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal is running the GNU nano 5.6.1 editor, editing a file named "protect.sh". The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install fail2ban
echo "Copy configuration files"
cp -R /vagrant/provision/server/protect/etc/* /etc
restorecon -vR /etc
echo "Start fail2ban service"
systemctl enable fail2ban
systemctl start fail2ban
```

Рисунок 24: Содержимое скрипта protect.sh

2.3.2 Создание скрипта protect.sh

- Установка fail2ban
- Копирование конфигурации в /etc
- Выполнен restorecon -vR /etc



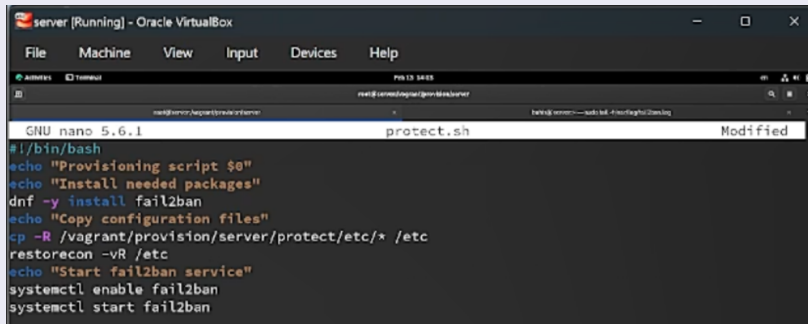
The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal is running the GNU nano 5.6.1 editor, editing a file named "protect.sh". The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install fail2ban
echo "Copy configuration files"
cp -R /vagrant/provision/server/protect/etc/* /etc
restorecon -vR /etc
echo "Start fail2ban service"
systemctl enable fail2ban
systemctl start fail2ban
```

Рисунок 24: Содержимое скрипта protect.sh

2.3.2 Создание скрипта protect.sh

- Установка fail2ban
- Копирование конфигурации в /etc
- Выполнен restorecon -vR /etc
- Включена и запущена служба



The screenshot shows a terminal window titled "server [Running] - Oracle VirtualBox". The terminal is running the GNU nano 5.6.1 editor, editing a file named "protect.sh". The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install fail2ban
echo "Copy configuration files"
cp -R /vagrant/provision/server/protect/etc/* /etc
restorecon -vR /etc
echo "Start fail2ban service"
systemctl enable fail2ban
systemctl start fail2ban
```

Рисунок 24: Содержимое скрипта protect.sh

2.3.3 Добавление provisioning в Vagrantfile

- Добавлен блок `server.vm.provision`

```
server.vm.provision "server protect",  
type: "shell",  
preserve_order: true,  
path: "provision/server/protect.sh"
```

Рисунок 25: Добавление блока provision в Vagrantfile

2.3.3 Добавление provisioning в Vagrantfile

- Добавлен блок `server.vm.provision`
- Указан путь `provision/server/protect.sh`

```
server.vm.provision "server protect",  
type: "shell",  
preserve_order: true,  
path: "provision/server/protect.sh"
```

Рисунок 25: Добавление блока provision в Vagrantfile

2.3.3 Добавление provisioning в Vagrantfile

- Добавлен блок `server.vm.provision`
- Указан путь `provision/server/protect.sh`
- Использован `preserve_order: true`

```
server.vm.provision "server protect",  
type: "shell",  
preserve_order: true,  
path: "provision/server/protect.sh"
```

Рисунок 25: Добавление блока provision в Vagrantfile

Раздел 3

3. Выводы

3.1 Выводы

- Установлен и настроен Fail2ban

3.1 Выводы

- Установлен и настроен Fail2ban
- Реализована защита SSH, HTTP и почтовых сервисов

3.1 Выводы

- Установлен и настроен Fail2ban
- Реализована защита SSH, HTTP и почтовых сервисов
- Проверена блокировка IP при превышении `maxretry`

3.1 Выводы

- Установлен и настроен Fail2ban
- Реализована защита SSH, HTTP и почтовых сервисов
- Проверена блокировка IP при превышении `maxretry`
- Выполнена ручная разблокировка

3.1 Выводы

- Установлен и настроен Fail2ban
- Реализована защита SSH, HTTP и почтовых сервисов
- Проверена блокировка IP при превышении `maxretry`
- Выполнена ручная разблокировка
- Настроен параметр `ignoreip`

3.1 Выводы

- Установлен и настроен Fail2ban
- Реализована защита SSH, HTTP и почтовых сервисов
- Проверена блокировка IP при превышении `maxretry`
- Выполнена ручная разблокировка
- Настроен параметр `ignoreip`
- Подтверждена корректная работа jail-модулей

3.1 Выводы

- Установлен и настроен Fail2ban
- Реализована защита SSH, HTTP и почтовых сервисов
- Проверена блокировка IP при превышении `maxretry`
- Выполнена ручная разблокировка
- Настроен параметр `ignoreip`
- Подтверждена корректная работа jail-модулей
- Настройка автоматизирована через provisioning

3.1 Выводы

- Установлен и настроен Fail2ban
- Реализована защита SSH, HTTP и почтовых сервисов
- Проверена блокировка IP при превышении `maxretry`
- Выполнена ручная разблокировка
- Настроен параметр `ignoreip`
- Подтверждена корректная работа jail-модулей
- Настройка автоматизирована через provisioning
- Обеспечена воспроизводимая базовая защита от brute force